

Package ‘methodical’

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Title Discovering genomic regions where methylation is strongly associated with transcriptional activity

Version 1.5.2

Description

DNA methylation is generally considered to be associated with transcriptional silencing. However, comprehensive, genome-wide investigation of this relationship requires the evaluation of potentially millions of correlation values between the methylation of individual genomic loci and expression of associated transcripts in a relatively large numbers of samples. Methodical makes this process quick and easy while keeping a low memory footprint. It also provides a novel method for identifying regions where a number of methylation sites are consistently strongly associated with transcriptional expression. In addition, Methodical enables housing DNA methylation data from diverse sources (e.g. WGBS, RRBS and methylation arrays) with a common framework, lifting over DNA methylation data between different genome builds and creating base-resolution plots of the association between DNA methylation and transcriptional activity at transcriptional start sites.

License GPL (>= 3)

BugReports <https://github.com/richardheery/methodical/issues>

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Description

DNA methylation is generally considered to be associated with transcriptional silencing. However, comprehensive, genome-wide investigation of this relationship requires the evaluation of potentially millions of correlation values between the methylation of individual genomic loci and expression of associated transcripts in a relatively large numbers of samples. Methodical makes this process quick and easy while keeping a low memory footprint. It also provides a novel method for identifying regions where a number of methylation sites are consistently strongly associated with transcriptional expression. In addition, Methodical enables housing DNA methylation data from diverse sources (e.g. WGBS, RRBS and methylation arrays) with a common framework, lifting over DNA methylation data between different genome builds and creating base-resolution plots of the association between DNA methylation and transcriptional activity at transcriptional start sites.

Author(s)

Richard Heery

See Also

Useful links:

- <https://github.com/richardheery/methodical>
- Report bugs at <https://github.com/richardheery/methodical/issues>

`.calculate_regions_intersections`

Calculate the number of bases in the intersection of two GRanges objects

Description

Calculate the number of bases in the intersection of two GRanges objects

Usage

```
.calculate_regions_intersections(
  gr1,
  gr2,
  ignore.strand = TRUE,
  overlap_measure = "absolute"
)
```

Arguments

<code>gr1</code>	A GRanges object
<code>gr2</code>	A GRanges object
<code>ignore.strand</code>	TRUE or FALSE indicating whether strand should be ignored when calculating intersections. Default is TRUE.

`overlap_measure`

One of "absolute", "proportion" or "jaccard" indicating whether to calculate the absolute size of the intersection in base pairs, the proportion base pairs of gr1 overlapping gr2 or the Jaccard index of the intersection in terms of base pairs. Default value is "absolute".

Value

An numeric value

<code>.chunk_regions</code>	<i>Split genomic regions into balanced chunks based on the number of methylation sites that they cover</i>
-----------------------------	--

Description

Split genomic regions into balanced chunks based on the number of methylation sites that they cover

Usage

```
.chunk_regions(  
  meth_rse,  
  genomic_regions,  
  max_sites_per_chunk = NULL,  
  ncores = 1  
)
```

Arguments

<code>meth_rse</code>	A RangedSummarizedExperiment with methylation values.
<code>genomic_regions</code>	A GRanges object.
<code>max_sites_per_chunk</code>	The maximum number of methylation sites to load into memory at once for each chunk.
<code>ncores</code>	The number of cores that will be used.

Value

A GRangesList where each GRanges object overlaps approximately the number of methylation sites given by `max_sites_per_chunk`

<code>.count_covered_bases</code>	<i>Calculate the number of unique bases covered by all regions in a GRanges object</i>
-----------------------------------	--

Description

Calculate the number of unique bases covered by all regions in a GRanges object

Usage

```
.count_covered_bases(gr)
```

Arguments

<code>gr</code>	A GRanges object
-----------------	------------------

Value

An numeric value

<code>.find_tmrs_single</code>	<i>Find TSS-Proximal Methylation-Controlled Regulatory Sites (TMRs)</i>
--------------------------------	---

Description

Find TSS-Proximal Methylation-Controlled Regulatory Sites (TMRs)

Usage

```
.find_tmrs_single(
  correlation_df,
  offset_length = 10,
  p_value_threshold = 0.05,
  smoothing_factor = 0.75,
  min_gapwidth = 150,
  min_meth_sites = 5
)
```

Arguments

<code>correlation_df</code>	A data.frame with correlation values between methylation sites and a transcript or a path to an RDS file containing such a data.frame as returned by <code>calculateMethSiteTranscriptCors</code> .
<code>offset_length</code>	Number of methylation sites added upstream and downstream of a central methylation site to form a window, resulting in a window size of $2 \times \text{offset_length} + 1$. Default value is 10.

`p_value_threshold` The `p_value` cutoff to use. Default value is 0.05.

`smoothing_factor` Smoothing factor for exponential moving average. Should be a value between 0 and 1 with higher values resulting in a greater degree of smoothing. Default is 0.75.

`min_gapwidth` Merge TMRs with the same direction separated by less than this number of base pairs. Default value is 150.

`min_meth_sites` Minimum number of methylation sites that TMRs can contain. Default value is 5.

Value

A GRanges object with the location of TMRs.

Examples

```
# Load methylation-transcript correlation results for TUBB6 gene
data("tubb6_cpg_meth_transcript_cors", package = "methodical")

# Find TMRs for
tubb6_tmrs <- methodical:::find_tmrs_single(correlation_df = tubb6_cpg_meth_transcript_cors)
print(tubb6_tmrs)
```

```
.summarize_chunk_methylation
```

Summarize methylation values for regions in a chunk

Description

Summarize methylation values for regions in a chunk

Usage

```
.summarize_chunk_methylation(
  chunk_regions,
  meth_rse,
  assay,
  col_summary_function,
  na.rm,
  ...
)
```

Arguments

chunk_regions	Chunk with genomic regions of interest.
meth_rse	A RangedSummarizedExperiment with methylation values.
assay	The assay from meth_rse to extract values from. Should be either an index or the name of an assay.
col_summary_function	A function that summarizes column values.
na.rm	TRUE or FALSE indicating whether to remove NA values when calculating summaries.
...	Additional arguments to be passed to col_summary_function.

Value

A function which returns a list with the summarized methylation values for regions in each sample.

.test_tmrs	<i>Find TMRs where smoothed methodical scores exceed thresholds</i>
------------	---

Description

Find TMRs where smoothed methodical scores exceed thresholds

Usage

```
.test_tmrs(
  meth_sites_gr,
  smoothed_methodical_scores,
  p_value_threshold = 0.05,
  tss_gr = NULL,
  transcript_id = NULL
)
```

Arguments

meth_sites_gr	A GRanges object with the location of methylation sites.
smoothed_methodical_scores	A numeric vector with the smoothed methodical scores associated with each methylation site.
p_value_threshold	The p_value cutoff to use. Default value is 0.05.
tss_gr	An optional GRanges object giving the location of the TSS meth_sites_gr is associated with.
transcript_id	Name of the transcript associated with the TSS.

Value

A GRanges object with the location of TMRs.

.tss_correlations	<i>Calculate meth site-transcript correlations for given TSS</i>
-------------------	--

Description

Calculate meth site-transcript correlations for given TSS

Usage

```
.tss_correlations(correlation_objects)
```

Arguments

correlation_objects

A list with a table of methylation values, expression values for transcripts, a GRangesList for the transcript and the name of the transcript.

Value

A data.frame with the correlation values

.tss_iterator	<i>Create an iterator function for use with bpiterate</i>
---------------	---

Description

Create an iterator function for use with bpiterate

Usage

```
.tss_iterator(  
  meth_values_chunk,  
  tss_region_indices_list,  
  transcript_values_list,  
  tss_gr_chunk_list,  
  cor_method,  
  add_distance_to_region,  
  min_number_complete_pairs,  
  results_dir  
)
```

Arguments

meth_values_chunk	A table with methylation values for current chunk
tss_region_indices_list	A list with the indices for methylation sites associated with each TSS.
transcript_values_list	A list with expression values for transcripts.
tss_gr_chunk_list	A list of GRanges with the TSS for the current chunk.
cor_method	Correlation method to use.
add_distance_to_region	TRUE or FALSE indicating whether to add distance to TSS.
min_number_complete_pairs	The minimum number of complete pairs required to return a p-value for a correlation.
results_dir	Location of results directory.

Value

An iterator function which returns a list with the parameters necessary for .tss_correlations.

annotateGRanges	<i>Annotate GRanges</i>
-----------------	-------------------------

Description

Annotate GRanges

Usage

```
annotateGRanges(
  genomic_regions,
  annotation_ranges,
  ignore.strand = TRUE,
  overlap_measure = "absolute"
)
```

Arguments

genomic_regions	A GRanges object to be annotated
annotation_ranges	A GRangesList object with GRanges for different features e.g. introns, exons, enhancers.

`ignore.strand` TRUE or FALSE indicating whether strand should be ignored when calculating intersections. Default is TRUE.

`overlap_measure` One of "absolute", "proportion" or "jaccard" indicating whether to calculate the absolute size of the intersection in base pairs, the proportion of base pairs of genomic_ranges overlapping one of the component GRanges of annotation_ranges, or the Jaccard index of the intersection in terms of base pairs. Default value is "absolute".

Value

A numeric vector with the overlap measure for genomic_regions with each type of region in annotation_ranges.

Examples

```
# Load annotation for CpG islands and repetitive DNA
data(hg38_cpg_islands, package = "methodical")
hg38_cpg_islands <- hg38_cpg_islands[hg38_cpg_islands$type == "hg38_cpg_islands"]
repeat_annotation_hg38 <- AnnotationHub::AnnotationHub()[["AH99003"]]

# Convert repeat_annotation_hg38 into a GRangesList
repeat_annotation_hg38 <- GRangesList(split(repeat_annotation_hg38, repeat_annotation_hg38$repClass))

# Calculate the proportion of base pairs in CpG islands overlapping different classes of repetitive elements
annotateGRanges(genomic_regions = hg38_cpg_islands, annotation_ranges = repeat_annotation_hg38, overlap_measure = "proportion")
```

annotatePlot	<i>Create a plot with genomic annotation for a plot of values at methylation sites.</i>
--------------	---

Description

Works with plots returned by `plotRegionValues()`, `plotMethSiteCorCoefs()` or `plotMethodicalScores`. Can combine the meth site values plot and genomic annotation together into a single plot or return the annotation plot separately.

Usage

```
annotatePlot(
  meth_site_plot,
  annotation_grl,
  reference_tss = FALSE,
  grl_colours = NULL,
  annotation_line_size = 5,
  ylab = "Genome Annotation",
  annotation_plot_proportion = 0.5,
```

```

    keep_meth_site_plot_legend = FALSE,
    annotation_plot_only = FALSE
  )

```

Arguments

meth_site_plot A plot of methylation site values (generally methylation level or correlation of methylation with transcription) around a TSS

annotation_grl A GRangesList object (or list coercible to a GRangesList) where each component GRanges gives the locations of different classes of regions to display. Each class of region will be given a separate colour in the plot, with regions ordered by the order of names(annotation_grl).

reference_tss TRUE or FALSE indicating whether to show distances on the X-axis relative to the TSS stored as an attribute tss_range of meth_site_plot. Alternatively, can provide a GRanges object with a single range for such a TSS site. In either case, will show the distance of methylation sites to the start of this region with methylation sites upstream relative to the reference_tss shown first. If FALSE (the default), the x-axis will instead show the start site coordinate of the methylation site. relative to the reference_tss shown first. If not, the x-axis will show the start site coordinate of the methylation site.

grl_colours An optional vector of colours used to display each of the GRanges making up annotation_grl. Must have same length as annotation_grl.

annotation_line_size Linewidth for annotation plot. Default is 5.

ylab The title to give the Y axis in the annotation plot. Default is "Genome Annotation".

annotation_plot_proportion A value giving the proportion of the height of the plot devoted to the annotation. Default is 0.5.

keep_meth_site_plot_legend TRUE or FALSE indicating whether to retain the legend of meth_site_plot, if it has one. Default value is FALSE.

annotation_plot_only TRUE or FALSE indicating whether to return only the annotation plot. Default is to combine meth_site_plot with the annotation.

Value

A ggplot object

Examples

```

# Get CpG islands from UCSC
data("hg38_cpg_islands", package = "methodical")
hg38_cpg_islands <- GRangesList(split(hg38_cpg_islands, hg38_cpg_islands$type))

# Load plot with CpG methylation correlation values for TUBB6
data("tubb6_correlation_plot", package = "methodical")

```

```
# Add positions of CpG islands to tubb6_correlation_plot
methodical::annotatePlot(tubb6_correlation_plot, annotation_grl = hg38_cpg_islands, annotation_plot_proportion =
```

```
calculateMethSiteTranscriptCors
```

Calculate correlation between expression of transcripts and methylation of sites surrounding their TSS

Description

Calculate correlation between expression of transcripts and methylation of sites surrounding their TSS

Usage

```
calculateMethSiteTranscriptCors(
  meth_rse,
  assay_number = 1,
  transcript_expression_table,
  samples_subset = NULL,
  tss_gr,
  tss_associated_gr,
  cor_method = "pearson",
  min_number_complete_pairs = 30,
  add_distance_to_region = TRUE,
  max_sites_per_chunk = NULL,
  BPPARAM = BiocParallel::SerialParam(),
  results_dir = NULL
)
```

Arguments

<code>meth_rse</code>	A <code>RangedSummarizedExperiment</code> for methylation sites.
<code>assay_number</code>	The assay from <code>meth_rse</code> to extract values from. Default is the first assay.
<code>transcript_expression_table</code>	A matrix or <code>data.frame</code> with the expression values for transcripts, where row names are transcript names and columns sample names. There should be a row corresponding to each transcript associated with each range in <code>tss_gr</code> . Names of samples must match those in <code>meth_rse</code> unless <code>samples_subset</code> provided.
<code>samples_subset</code>	Sample names used to subset <code>meth_rse</code> and <code>transcript_expression_table</code> . Provided samples must be found in both <code>meth_rse</code> and <code>transcript_expression_table</code> . Default is to use all samples in <code>meth_rse</code> and <code>transcript_expression_table</code> .
<code>tss_gr</code>	A <code>GRanges</code> object with the locations of transcription start sites. Names of regions cannot contain any duplicates and should and match those of <code>tss_associated_gr</code> and be present in <code>transcript_expression</code> table.

tss_associated_gr	A GRanges object with the locations of regions associated with each transcription start site. Names of regions cannot contain any duplicates and should match those of tss_gr and be present in transcript_expression table.
cor_method	A character string indicating which correlation coefficient is to be computed. One of either "pearson" or "spearman" or their abbreviations.
min_number_complete_pairs	The minimum number of complete pairs required to return a p-value for a correlation. Correlations with less than this number are given a p-value of NaN. Default value is 30.
add_distance_to_region	TRUE or FALSE indicating whether to add the distance of methylation sites to the TSS. Default value is TRUE. Setting to FALSE will roughly half the total running time.
max_sites_per_chunk	The approximate maximum number of methylation sites to try to load into memory at once. The actual number loaded may vary depending on the number of methylation sites overlapping each region, but so long as the size of any individual regions is not enormous ($\geq$ several MB), it should vary only very slightly. Some experimentation may be needed to choose an optimal value as low values will result in increased running time, while high values will result in a large memory footprint without much improvement in running time. Default is $\text{floor}(62500000/\text{ncol}(\text{meth_rse}))$, resulting in each chunk requiring approximately 500 MB of RAM.
BPPARAM	A BiocParallelParam object for parallel processing. Defaults to <code>BiocParallel::SerialParam()</code> .
results_dir	An optional path to a directory to save results as RDS files. There will be one RDS file for each transcript. If not provided, returns the results as a list.

Value

If results_dir is NULL, a list of data.frames with the correlation of methylation sites surrounding a specified genomic region with a given feature, p-values and adjusted q-values for the correlations. Distance of the methylation sites upstream or downstream to the center of the region is also provided. If results_dir is provided, instead returns a list with the paths to the RDS files with the results.

Examples

```
# Load TUBB6 TSS GRanges, RangedSummarizedExperiment with methylation values for CpGs around TUBB6 TSS and TUBB6 tra
data(tubb6_tss, package = "methodical")
data(tubb6_meth_rse, package = "methodical")
tubb6_meth_rse <- eval(tubb6_meth_rse)
data(tubb6_transcript_counts, package = "methodical")

# Calculate correlation values between methylation values and transcript values for TUBB6
tubb6_cpg_meth_transcript_cors <- methodical::calculateMethSiteTranscriptCors(meth_rse = tubb6_meth_rse,
  transcript_expression_table = tubb6_transcript_counts, tss_gr = tubb6_tss,
  tss_associated_gr = methodical::expand_granges(tubb6_tss, upstream = 5000, downstream = 5000))
head(tubb6_cpg_meth_transcript_cors$ENST00000591909)
```

```
calculateRegionMethylationTranscriptCors
```

Calculate the correlation values between the methylation of genomic regions and the expression of associated transcripts

Description

Calculate the correlation values between the methylation of genomic regions and the expression of associated transcripts

Usage

```
calculateRegionMethylationTranscriptCors(
  meth_rse,
  assay = 1,
  transcript_expression_table,
  samples_subset = NULL,
  genomic_regions,
  genomic_region_names = NULL,
  genomic_region_transcripts = NULL,
  genomic_region_methylation = NULL,
  cor_method = "pearson",
  p_adjust_method = "BH",
  region_methylation_summary_function = colMeans,
  BPPARAM = BiocParallel::SerialParam(),
  ...
)
```

Arguments

meth_rse	A RangedSummarizedExperiment with methylation values for CpG sites which will be used to calculate methylation values for genomic_regions. There must be at least 3 samples in common between meth_rse and transcript_expression_table.
assay	The assay from meth_rse to extract values from. Should be either an index or the name of an assay. Default is the first assay.
transcript_expression_table	A table with the expression values for different transcripts in different samples. Row names should give be the transcript name and column names should be the name of samples.
samples_subset	Optional sample names used to subset meth_rse and transcript_expression_table. Provided samples must be found in both meth_rse and transcript_expression_table. Default is to use all samples in meth_rse and transcript_expression_table.
genomic_regions	A GRanges object.

<code>genomic_region_names</code>	A character vector of unique names to assign <code>genomic_regions</code> in the output table. Defaults to <code>names(genomic_regions)</code> if present or otherwise converts regions to character strings (e.g. "chr:1000-2000") to use as names.
<code>genomic_region_transcripts</code>	Names of transcripts associated with each region in <code>genomic_regions</code> . If not provided, attempts to use <code>genomic_regions\$transcript_id</code> . All transcripts must be present in <code>transcript_expression_table</code> .
<code>genomic_region_methylation</code>	Optional preprovided table with methylation values for <code>genomic_regions</code> such as created using <code>summarizeRegionMethylation()</code> . Table will be created if it is not provided which will increase running time. Row names should match <code>genomic_region_names</code> and column names should match those of <code>transcript_expression_table</code>
<code>cor_method</code>	A character string indicating which correlation coefficient is to be computed. One of either "pearson" or "spearman" or their abbreviations.
<code>p_adjust_method</code>	Method used to adjust p-values. Same as the methods from <code>p.adjust.methods</code> . Default is Benjamini-Hochberg.
<code>region_methylation_summary_function</code>	A function that summarizes column values. Default is <code>colMeans</code> .
<code>BPPARAM</code>	A <code>BiocParallelParam</code> object for parallel processing. Defaults to <code>BiocParallel::SerialParam()</code> .
<code>...</code>	Additional arguments to be passed to <code>summary_function</code> .

Value

A `data.frame` with the correlation values between the methylation of genomic regions and expression of transcripts associated with them

Examples

```
# Load TUBB6 TMRs, RangedSummarizedExperiment with methylation values for CpGs around TUBB6 TSS and TUBB6 transcript
data(tubb6_tmrs, package = "methodical")
data(tubb6_meth_rse, package = "methodical")
tubb6_meth_rse <- eval(tubb6_meth_rse)
data(tubb6_transcript_counts, package = "methodical")

# Calculate correlation values between TMRs identified for TUBB6 and transcript expression
tubb6_tmrs_transcript_cors <- methodical::calculateRegionMethylationTranscriptCors(
  meth_rse = tubb6_meth_rse, transcript_expression_table = tubb6_transcript_counts,
  genomic_regions = tubb6_tmrs, genomic_region_names = tubb6_tmrs$tmr_name)
tubb6_tmrs_transcript_cors
```

`calculateSmoothedMethodicalScores`*Calculate methodical score and smooth it using a exponential weighted moving average*

Description

Calculate methodical score and smooth it using a exponential weighted moving average

Usage

```
calculateSmoothedMethodicalScores(  
  correlation_df,  
  offset_length = 10,  
  smoothing_factor = 0.75  
)
```

Arguments

<code>correlation_df</code>	A data.frame with correlation values between methylation sites and a transcript as returned by <code>calculateMethSiteTranscriptCors</code> .
<code>offset_length</code>	Number of methylation sites added upstream and downstream of a central methylation site to form a window, resulting in a window size of $2 \times \text{offset_length} + 1$. Default value is 10.
<code>smoothing_factor</code>	Smoothing factor for exponential moving average. Should be a value between 0 and 1 with higher values resulting in a greater degree of smoothing. Default is 0.75.

Value

A GRanges object

Examples

```
# Load data.frame with CpG methylation-transcription correlation results for TUBB6  
data("tubb6_cpg_meth_transcript_cors", package = "methodical")  
  
# Calculate smoothed Methodical scores from correlation values  
smoothed_methodical_scores <- methodical::calculateSmoothedMethodicalScores(tubb6_cpg_meth_transcript_cors)
```

correct_correlation_pvalues

Correct p-values for a list of methylation-transcription correlations results

Description

Correct p-values for a list of methylation-transcription correlations results

Usage

```
correct_correlation_pvalues(correlation_list, p_adjust_method = "fdr")
```

Arguments

correlation_list

A list of data.frames with correlation values between methylation sites and a transcript as returned by calculateMethSiteTranscriptCors.

p_adjust_method

The method to use for p-value adjustment. Should be one of the methods in p.adjust.methods. Default is "fdr".

Value

A list identical to correlation_list except with p-values corrected using the indicated method.

createRandomRegions *Create a GRanges with random regions from a genome*

Description

Can constrain the random regions so that they do not overlap each other and/or an optional set of masked regions. Random regions which do meet these constraints will be discarded and new ones generated until the desired number of regions has been reached or a maximum allowed number of attempts has been made. After the maximum number of allowed attempts, the created random regions meeting the constraints up to that point will be returned. Any random regions that are out-of-bounds relative to their sequence length are trimmed before being returned.

Usage

```
createRandomRegions(
  genome,
  n_regions = 1000,
  region_widths = 1000,
  sequence_names = NULL,
  all_sequence_names_equally_likely = FALSE,
```

```

    stranded = FALSE,
    masked_regions = NULL,
    allow_overlapping_regions = FALSE,
    ignore.strand = TRUE,
    max_tries = 100
  )

```

Arguments

genome	A BSgenome object.
n_regions	Number of random regions to create. Default is 1000.
region_widths	The widths of the random regions. Widths cannot be negative. Can be just a single value if all regions are to have the same widths. Default is 1000.
stranded	TRUE or FALSE indicating if created regions should have a strand randomly assigned. Default is FALSE, indicating to make unstranded regions.
masked_regions	An optional GRanges object which random regions will not be allowed to overlap.
allow_overlapping_regions	TRUE or FALSE indicating if created random regions should be allowed to overlap. Default is FALSE.
ignore.strand	TRUE or FALSE indicating whether strand should be ignored when identifying overlaps between random regions with each other or with masked_regions. Only relevant if stranded is TRUE and either allow_overlapping_regions is FALSE or masked_regions is provided. Default is TRUE.
max_tries	The maximum number of attempts to make to find non-overlapping regions which do not overlap masked_regions. Default value is 100.
sequences	The names of sequences to create random regions on. Default is to use all sequences in the genome.
all_sequences_equally_likely	TRUE or FALSE indicating if the probability of creating random regions on a sequence should be the same for each sequence. Default is FALSE, indicating to make the probability proportional to a sequences length.

Value

A GRanges object

Examples

```

# Set random seed
set.seed(123)

# Create 10,000 random non-overlapping regions with width 1,000 for hg38
random_regions <- methodical::createRandomRegions(genome = "BSgenome.Hsapiens.UCSC.hg38", n_regions = 10000)
head(random_regions)

```

expand_granges	<i>Expand GRanges</i>
----------------	-----------------------

Description

Expand ranges in a GRanges object upstream and downstream by specified numbers of bases, taking account of strand. Unstranded ranges are treated like they on the "+" strand. If any of the resulting ranges are out-of-bounds given the seqinfo of genomic_regions, they will be trimmed using trim().

Usage

```
expand_granges(genomic_regions, upstream = 0, downstream = 0)
```

Arguments

genomic_regions	A GRanges object
upstream	Number of bases to add upstream of each region in genomic_regions. Must be numeric vector of length 1 or else equal to the length of genomic_regions. Default value is 0. Negative values result in upstream end of regions being shortened, however the width of the resulting regions cannot be less than zero.
downstream	Number of bases to add downstream of each region in genomic_regions. Negative values result in downstream end of regions being shortened. Must be numeric vector of length 1 or else equal to the length of genomic_regions. Default value is 0. Negative values result in upstream end of regions being shortened, however the width of the resulting regions cannot be less than zero.

Value

A GRanges object

Examples

```
data(tubb6_tss, package = "methodical")
tubb6_tss
methodical::expand_granges(tubb6_tss, upstream = 5000, downstream = 5000)
```

extractGRangesMethSiteValues	<i>Extract values for methylation sites overlapping genomic regions from a methylation RSE.</i>
------------------------------	---

Description

Extract values for methylation sites overlapping genomic regions from a methylation RSE.

Usage

```
extractGRangesMethSiteValues(
  meth_rse,
  genomic_regions = NULL,
  samples_subset = NULL,
  assay_number = 1
)
```

Arguments

meth_rse A RangedSummarizedExperiment for methylation data.

genomic_regions A GRanges object. If set to NULL, returns all methylation sites in meth_rse

samples_subset Optional sample names used to subset meth_rse.

assay_number The assay from meth_rse to extract values from. Default is the first assay.

Value

A data.frame with the methylation site values for all sites in meth_rse which overlap genomic_ranges. Row names are the coordinates of the sites as a character vector.

Examples

```
# Load sample RangedSummarizedExperiment with CpG methylation data
data(tubb6_meth_rse, package = "methodical")
tubb6_meth_rse <- eval(tubb6_meth_rse)

# Create a sample GRanges object to use
test_region <- GRanges("chr18:12305000-12310000")

# Get methylation values for CpG sites overlapping HDAC1 gene
test_region_methylation <- methodical::extractGRangesMethSiteValues(meth_rse = tubb6_meth_rse, genomic_regions =
```

```
extractMethSitesFromGenome
```

Create a GRanges with methylation sites of interest from a BSgenome.

Description

Create a GRanges with methylation sites of interest from a BSgenome.

Usage

```
extractMethSitesFromGenome(
  genome,
  pattern = "CG",
  plus_strand_only = TRUE,
  meth_site_position = 1,
  standard_sequences_only = TRUE
)
```

Arguments

genome	A BSgenome object (or the name of an installed one) or a DNASTringSet with names indicating the sequences.
pattern	A pattern to match in genome. Default is "CG".
plus_strand_only	TRUE or FALSE indicating whether to only return matches on "+" strand, avoiding returning duplicate hits for palindromic sequences e.g. CG. Not relevant if genome is a DNASTringSet. Default is TRUE.
meth_site_position	Which position in the pattern corresponds to the methylation site of interest. Default is the first position.
standard_sequences_only	TRUE or FALSE indicating whether to only return sites on standard sequences (those without "-" in their names). Default is TRUE.

Value

A GRanges object with genomic regions matching the pattern.

Examples

```
# Get human CpG sites for hg38 genome build
hg38_cpgs <- methodical::extractMethSitesFromGenome("BSgenome.Hsapiens.UCSC.hg38")
head(hg38_cpgs)

# Find CHG sites in Arabidopsis thaliana
arabidopsis_cphpgs <- methodical::extractMethSitesFromGenome("BSgenome.Athaliana.TAIR.TAIR9", pattern = "CHG")
head(arabidopsis_cphpgs)
```

findTMRs

Find TSS-Proximal Methylation-Controlled Regulatory Sites (TMRs)

Description

Find TSS-Proximal Methylation-Controlled Regulatory Sites (TMRs)

Usage

```
findTMRs(  
  correlation_list,  
  offset_length = 10,  
  p_adjust_method = "fdr",  
  p_value_threshold = 0.05,  
  smoothing_factor = 0.75,  
  min_gapwidth = 150,  
  min_meth_sites = 5,  
  BPPARAM = BiocParallel::SerialParam()  
)
```

Arguments

- | | |
|--------------------------------|--|
| <code>correlation_list</code> | A list of data.frames with correlation values between methylation sites and a transcript as returned by <code>calculateMethSiteTranscriptCors</code> . |
| <code>offset_length</code> | Number of methylation sites added upstream and downstream of a central methylation site to form a window, resulting in a window size of $2 * \text{offset_length} + 1$. Default value is 10. |
| <code>p_adjust_method</code> | The method to use for p-value adjustment. Should be one of the methods in <code>p.adjust.methods</code> . Default is "fdr". |
| <code>p_value_threshold</code> | The p_value cutoff to use (after correcting p-values with <code>p_adjust_method</code>). Default value is 0.05. |
| <code>smoothing_factor</code> | Smoothing factor for exponential moving average. Should be a value between 0 and 1 with higher values resulting in a greater degree of smoothing. Default is 0.75. |
| <code>min_gapwidth</code> | Merge TMRs with the same direction separated by less than this number of base pairs. Default value is 150. |
| <code>min_meth_sites</code> | Minimum number of methylation sites that TMRs can contain. Default value is 5. |
| <code>BPPARAM</code> | A <code>BiocParallelParam</code> object for parallel processing. Defaults to <code>BiocParallel::SerialParam()</code> . |

Value

A `GRanges` object with the location of TMRs.

hg38_cpgs_subset	<i>hg38_cpgs_subset</i>
------------------	-------------------------

Description

All the CpG sites within the first one million base pairs of chromosome 1.

Usage

```
hg38_cpgs_subset
```

Format

A GRanges object.

kallistoIndex	<i>Create an index file for running Kallisto</i>
---------------	--

Description

Create an index file for running Kallisto

Usage

```
kallistoIndex(
  path_to_kallisto,
  transcripts_fasta,
  index_name = "kallisto_index.idx"
)
```

Arguments

path_to_kallisto	Path to kallisto executable
transcripts_fasta	Path to a fasta file for the transcripts to be quantified.
index_name	Name to give the created index file. Default is "kallisto_index.idx".

Value

Invisibly returns TRUE.

Examples

```
## Not run:
# Download transcripts FASTA from Gencode
download.file("https://ftp.ebi.ac.uk/pub/databases/gencode/Gencode_human/release_44/gencode.v44.transcripts.fa.gz")

# Locate the kallisto executable (provided that it is on the path)
kallisto_path <- system2("which", args = "kallisto", stdout = TRUE)

# Create transcripts index for use with Kallisto
methodical::kallistoIndex(kallisto_path, transcripts_fasta = "gencode.v44.transcripts.fa.gz")

## End(Not run)
```

kallistoQuantify	<i>Run kallisto on a set of FASTQ files and merge the results</i>
------------------	---

Description

Run kallisto on a set of FASTQ files and merge the results

Usage

```
kallistoQuantify(
  path_to_kallisto,
  kallisto_index,
  forward_fastq_files,
  reverse_fastq_files,
  sample_names,
  output_directory,
  merged_output_prefix = "kallisto_transcript",
  messages_file = "",
  ncores = 1,
  number_bootstraps = 100
)
```

Arguments

path_to_kallisto	Path to kallisto executable
kallisto_index	Path to a kallisto index
forward_fastq_files	A vector with the paths to forward FASTQ files. Each file should correspond to the file at the same position in reverse_fastq_files.
reverse_fastq_files	A vector with the paths to reverse FASTQ files. Each file should correspond to the file at the same position in forward_fastq_files.

sample_names	A vector with the names of samples for each pair of samples from forward_fastq_files and reverse_fastq_files
output_directory	The name of the directory to save results in. Will be created if it doesn't exist.
merged_output_prefix	Prefix to use for names of merged output files for counts and TPM which take the form merged_output_prefix_counts_merged.tsv.gz and merged_output_prefix_tpm_merged.tsv.gz. Default prefix is "kallisto_transcript" i.e. default output merged output files are kallisto_transcript_counts_merged.tsv.gz and kallisto_transcript_tpm_merged.tsv.gz.
messages_file	Name of file to save kallisto run messages. If no file name given, information is printed to stdout.
ncores	The number of cores to use. Default is 1.
number_bootstraps	The number of bootstrap samples. Default is 100.

Value

The path to the merged counts table.

liftoverMethRSE	<i>Liftover rowRanges of a RangedSummarizedExperiment for methylation data from one genome build to another</i>
-----------------	---

Description

Removes methylation sites which cannot be mapped to the target genome build and those which result in many-to-one mappings. Also removes one-to-many mappings by default and can remove sites which do not map to allowed regions in the target genome e.g. CpG sites.

Usage

```
liftoverMethRSE(
  meth_rse,
  chain,
  remove_one_to_many_mapping = TRUE,
  permitted_target_regions = NULL,
  seqlevels = NULL
)
```

Arguments

meth_rse	A RangedSummarizedExperiment for methylation data
chain	A "Chain" object to be used with rtracklayer::liftOver
remove_one_to_many_mapping	TRUE or FALSE indicating whether to remove regions in the source genome which map to multiple regions in the target genome. Default is TRUE.

permitted_target_regions An optional GRanges object used to filter the rowRanges by overlaps after liftover, for example CpG sites from the target genome. Any regions which do not overlap permitted_target_regions will be removed. GRangesList to GRanges if all remaining source regions can be uniquely mapped to the target genome.

seqlevels An optional character vector giving the order to use for seqlevels of the rowRanges of the returned RangedSummarizedExperiment.

Value

A RangedSummarizedExperiment with rowRanges lifted over to the genome build indicated by chain.

Examples

```
# Load sample RangedSummarizedExperiment with CpG methylation data
data(tubb6_meth_rse, package = "methodical")
tubb6_meth_rse <- eval(tubb6_meth_rse)

# Get CpG sites for hg19
hg19_cpgs <- methodical::extractMethSitesFromGenome("BSgenome.Hsapiens.UCSC.hg19")

# Get liftover chain for mapping hg38 to hg19
library(AnnotationHub)
ah <- AnnotationHub()
chain <- ah[["AH14108"]]

# Liftover tubb6_meth_rse from hg38 to hg19, keeping only sites that were mapped to CpG sites in hg19
tubb6_meth_rse_hg19 <- methodical::liftoverMethRSE(tubb6_meth_rse, chain = chain,
  permitted_target_regions = hg19_cpgs)
```

maskRangesInRSE	<i>Mask regions in a ranged summarized experiment</i>
-----------------	---

Description

Mask regions in a ranged summarized experiment

Usage

```
maskRangesInRSE(rse, mask_ranges, assay_number = 1)
```

Arguments

rse A RangedSummarizedExperiment.

mask_ranges Either a GRanges with regions to be masked in all samples (e.g. repetitive sequences) or a GRangesList object with different regions to mask in each sample (e.g. mutations). If using a GRangesList object, names of the list elements should be the names of samples in rse.

assay_number Assay to perform masking. Default is first assay

Value

A RangedSummarizedExperiment with the regions present in mask_ranges masked

Examples

```
# Load sample RangedSummarizedExperiment with CpG methylation data
data(tubb6_meth_rse, package = "methodical")
tubb6_meth_rse <- eval(tubb6_meth_rse)

# Create a sample GRanges object to use to mask tubb6_meth_rse
mask_ranges <- GRanges("chr18:12305000-12310000")

# Mask regions in tubb6_meth_rse
tubb6_meth_rse_masked <- methodical::maskRangesInRSE(tubb6_meth_rse, mask_ranges)

# Count the number of NA values before and after masking
sum(is.na(assay(tubb6_meth_rse)))
sum(is.na(assay(tubb6_meth_rse_masked)))
```

plotMethodicalScores	<i>Create plot of Methodical score values for methylation sites around a TSS</i>
----------------------	--

Description

Create plot of Methodical score values for methylation sites around a TSS

Usage

```
plotMethodicalScores(
  genomic_region_values,
  reference_tss = NULL,
  p_value_threshold = 0.005,
  smooth_scores = TRUE,
  offset_length = 10,
  smoothing_factor = 0.75,
  smoothed_curve_colour = "black",
  linewidth = 1,
  curve_alpha = 0.75,
  title = NULL,
  xlabel = "Genomic Position",
  low_colour = "#7B5C90",
  high_colour = "#BFAB25"
)
```

Arguments

genomic_region_values	A data.frame with correlation values for methylation sites. There should be one column called "cor". and another called "p_val" which are used to calculate the Methodical score. row.names should be the names of methylation sites and all methylation sites must be located on the same sequence.
reference_tss	An optional GRanges object with a single range. If provided, the x-axis will show the distance of methylation sites to the start of this region with methylation sites upstream. relative to the reference_tss shown first. If not, the x-axis will show the start site coordinate of the methylation site.
p_value_threshold	The p-value threshold used to identify TMRs. Default value is 0.005. Set to NULL to turn off significance thresholds.
smooth_scores	TRUE or FALSE indicating whether to display a curve of smoothed Methodical scores on top of the plot. Default is TRUE.
offset_length	Offset length to be supplied to calculateSmoothedMethodicalScores. Default is 10.
smoothing_factor	Smoothing factor to be provided to calculateSmoothedMethodicalScores. Default is 0.75.
smoothed_curve_colour	Colour of the smoothed curve. Default is "black".
linewidth	Line width of the smoothed curve. Default value is 1.
curve_alpha	Alpha value for the curve. Default value is 0.75.
title	Title of the plot. Default is no title.
xlabel	Label for the X axis in the plot. Default is "Genomic Position".
low_colour	Colour to use for low values. Default value is "#7B5C90".
high_colour	Colour to use for high values. Default value is "#BFAB25".

Value

A ggplot object

Examples

```
# Load methylation-transcript correlation results for TUBB6 gene
data("tubb6_cpg_meth_transcript_cors", package = "methodical")

# Calculate and plot Methodical scores from correlation values
methodical::plotMethodicalScores(tubb6_cpg_meth_transcript_cors, reference_tss = attributes(tubb6_cpg_meth_tran
```

`plotMethSiteCorCoefs` *Plot the correlation coefficients for methylation sites within a region and an associated feature of interest*

Description

Plot the correlation coefficients for methylation sites within a region and an associated feature of interest

Usage

```
plotMethSiteCorCoefs(
  meth_site_cor_values,
  reference_tss = FALSE,
  title = NULL,
  xlabel = NULL,
  ylabel = "Correlation Coefficient",
  value_colours = c("#7B5C90", "#bfab25"),
  reverse_x_axis = FALSE
)
```

Arguments

<code>meth_site_cor_values</code>	A data.frame with correlation values associated with methylation sites, such as returned by <code>calculateMethSiteTranscriptCors</code> . There should be one column called <code>meth_site</code> giving the coordinates of methylation sites in character format and another column called <code>cor</code> giving the correlation between the methylation values of the methylation sites and a feature of interest. All methylation sites must be located on the same sequence.
<code>reference_tss</code>	TRUE or FALSE indicating whether to show distances on the X-axis relative to the TSS stored as an attribute <code>tss_range</code> of <code>meth_site_cor_values</code> . Alternatively, can provide a <code>GRanges</code> object with a single range for such a TSS site. In either case, will show the distance of methylation sites to the start of this region with methylation sites upstream relative to the <code>reference_tss</code> shown first. If FALSE (the default), the x-axis will instead show the start site coordinate of the methylation site.
<code>title</code>	Title of the plot. Default is no title.
<code>xlabel</code>	Label for the X axis in the plot. Defaults to "Distance to TSS" if <code>reference_tss</code> is used or "seqname position" where <code>seqname</code> is the name of the relevant sequence.
<code>ylabel</code>	Label for the Y axis in the plot. Default is "Correlation Coefficient".
<code>value_colours</code>	A vector with two colours to use, the first for low values and the second for high values. Defaults are <code>c("#7B5C90", "#bfab25")</code> .
<code>reverse_x_axis</code>	TRUE or FALSE indicating whether x-axis should be reversed, for example if plotting a region on the reverse strand so that left side of plot corresponds to upstream.

Value

A ggplot object

Examples

```
# Load methylation-transcript correlation results for TUBB6 gene
data("tubb6_cpg_meth_transcript_cors", package = "methodical")

# Plot methylation-transcript correlation values around TUBB6 TSS
methodical::plotMethSiteCorCoefs(tubb6_cpg_meth_transcript_cors, ylabel = "Spearman Correlation")

# Create same plot but showing the distance to the TUBB6 TSS on the x-axis
methodical::plotMethSiteCorCoefs(tubb6_cpg_meth_transcript_cors,
  ylabel = "Spearman Correlation", reference_tss = attributes(tubb6_cpg_meth_transcript_cors)$tss_range)
```

plotRegionValues	<i>Create a scatter plot with smoothed curve for values along adjacent loci in a genomic region</i>
------------------	---

Description

Create a scatter plot with smoothed curve for values along adjacent loci in a genomic region

Usage

```
plotRegionValues(
  genomic_region_values,
  sample_name = NULL,
  reference_tss = FALSE,
  geom_point_params = list(),
  geom_smooth_params = list(),
  title = NULL,
  xlabel = NULL,
  ylabel = "Genomic Region Value",
  value_colours = c("#53868B", "#CD2626"),
  reverse_x_axis = FALSE
)
```

Arguments

genomic_region_values	A data.frame with values associated with genomic regions. Row names must be the coordinates of genomic regions in character format (e.g chr1:1000-2000) and all regions must be located on the same sequence. The position of the first base in each region is used as the x-axis coordinate for the plot.
sample_name	Name of column in genomic_region_values to plot. Defaults to first column if none provided.

reference_tss	TRUE or FALSE indicating whether to show distances on the X-axis relative to the TSS stored as an attribute <code>tss_range</code> of <code>genomic_region_values</code> . Alternatively, can provide a <code>GRanges</code> object with a single range for such a TSS site. In either case, will show the distance of genomic regions to the start of this region with genomic regions upstream relative to the reference_tss shown first. If FALSE (the default), the x-axis will instead show the start site coordinate of the genomic region.
geom_point_params	An optional list to explicitly set values of parameters to use with <code>geom_point()</code> . Use <code>list(alpha = 0)</code> to make points invisible.
geom_smooth_params	An optional list to explicitly set values of parameters to use with <code>geom_smooth()</code> . Use <code>list(alpha = 0)</code> to make line invisible.
title	Title of the plot. Default is no title.
xlabel	Label for the X axis in the plot. Defaults to "Distance to TSS" if <code>reference_tss</code> is used or "seqname position" where <code>seqname</code> is the name of the relevant sequence.
ylabel	Label for the Y axis in the plot. Default is "Genomic Region Value".
value_colours	A vector with two colours to use, the first for low values and the second for high values. Defaults are <code>c("#53868B", "#CD2626")</code> .
reverse_x_axis	TRUE or FALSE indicating whether x-axis should be reversed, for example if plotting a region on the reverse strand so that left side of plot corresponds to upstream.

Value

A ggplot object

Examples

```
# Load methylation-values around the TUBB6 TSS
data("tubb6_meth_rse", package = "methodical")
tubb6_meth_rse <- eval(tubb6_meth_rse)

# Extract methylation values from tubb6_meth_rse
tubb6_methylation_values = methodical::extractGRangesMethSiteValues(meth_rse = tubb6_meth_rse)

# Plot methylation values around TUBB6 TSS
methodical::plotRegionValues(tubb6_methylation_values, sample_name = "N1", ylabel = "Methylation Value")

# Create same plot but showing the distance to the TUBB6 TSS on the x-axis
data("tubb6_tss", package = "methodical")
methodical::plotRegionValues(tubb6_methylation_values, sample_name = "N1",
  reference_tss = tubb6_tss, ylabel = "Methylation Value")
```


plotTMRs

*Add TMRs to a methylation site value plot***Description**

Add TMRs to a methylation site value plot

Usage

```
plotTMRs(
  meth_site_plot,
  tmrs_gr,
  reference_tss = NULL,
  transcript_id = NULL,
  tmr_colours = c("#A28CB1", "#D2C465"),
  linewidth = 5
)
```

Arguments

meth_site_plot	A plot of Value around a TSS.
tmrs_gr	A GRanges object giving the position of TMRs.
reference_tss	An optional GRanges object with a single range. If provided, the x-axis will show the distance of methylation sites to the start of this region with methylation sites upstream relative to the reference_tss shown first. If not, the x-axis will show the start site coordinate of the methylation site.
transcript_id	An optional transcript ID. If provided, will attempt to filter tmrs_gr and reference_tss using a metadata column called transcript_id with a value identical to the provided one.
tmr_colours	A vector with colours to use for negative and positive TMRs. Defaults to "#7B5C90" for negative and "#BFAB25" for positive TMRs.
linewidth	A numeric value to be provided as linewidth for geom_segment().

Value

A ggplot object

Examples

```
# Load methylation-transcript correlation results for TUBB6 gene
data("tubb6_cpg_meth_transcript_cors", package = "methodical")

# Plot methylation-transcript correlation values around TUBB6 TSS
tubb6_correlation_plot <- methodical::plotMethSiteCorCoefs(tubb6_cpg_meth_transcript_cors, ylabel = "Spearman Co")

# Find TMRs for TUBB6
```

```
tubb6_tmrs <- findTMRs(correlation_list = list(ENST00000591909 = tubb6_cpg_meth_transcript_cors))

# Plot TMRs on top of tubb6_correlation_plot
methodical::plotTMRs(tubb6_correlation_plot, tmrs_gr = tubb6_tmrs)
```

`rangesRelativeToTSS` *Find locations of genomic regions relative to transcription start sites.*

Description

Find locations of genomic regions relative to transcription start sites.

Usage

```
rangesRelativeToTSS(genomic_regions, tss_gr)
```

Arguments

`genomic_regions` A GRanges object.

`tss_gr` A GRanges object with transcription start sites. Each range should have width 1. Upstream and downstream are relative to strand of `tss_gr`.

Value

A GRanges object where all regions have "relative" as the sequence names and ranges are the location of TMRs relative to the TSS.

Examples

```
# Create query and subject GRanges
genomic_regions <- GenomicRanges::GRanges(c("chr1:100-1000:+", "chr1:2000-3000:-"))
tss_gr <- GenomicRanges::GRanges(c("chr1:1500:+", "chr1:4000:-"))

# Calculate distances between query and subject
methodical::rangesRelativeToTSS(genomic_regions, tss_gr)
```

rapidCorTest	<i>Rapidly calculate the correlation and the significance of pairs of columns from two data.frames</i>
--------------	--

Description

Rapidly calculate the correlation and the significance of pairs of columns from two data.frames

Usage

```
rapidCorTest(
  table1,
  table2,
  cor_method = "pearson",
  table1_name = "table1",
  table2_name = "table2",
  p_adjust_method = "BH",
  n_covariates = 0,
  min_number_complete_pairs = 30
)
```

Arguments

table1	A data.frame
table2	A data.frame
cor_method	A character string indicating which correlation coefficient is to be computed. One of either "pearson" or "spearman" or their abbreviations.
table1_name	Name to give the column giving the name of features in table1. Default is "table1".
table2_name	Name to give the column giving the name of features in table2. Default is "table2".
p_adjust_method	Method used to adjust p-values. Same as the methods from p.adjust.methods. Default is Benjamini-Hochberg. Setting to "none" will result in no adjusted p-values being calculated.
n_covariates	Number of covariates if calculating partial correlations. Defaults to 0.
min_number_complete_pairs	The minimum number of complete pairs required to return a p-value for a correlation. Correlations with less than this number are given a p-value of NaN. Default value is 30.

Value

A data.frame with the correlation and its significance for all pairs consisting of a variable from table1 and a variable from table2.

Examples

```
# Divide mtcars into two tables
table1 <- mtcars[, 1:5]
table2 <- mtcars[, 6:11]

# Calculate correlation between table1 and table2
cor_results <- methodical::rapidCorTest(table1, table2, cor_method = "spearman",
  table1_name = "feature1", table2_name = "feature2")
head(cor_results)
```

sampleMethSites	<i>Randomly sample sites from a methylation RSE.</i>
-----------------	--

Description

Randomly sample sites from a methylation RSE.

Usage

```
sampleMethSites(
  meth_rse,
  n_sites = 1000,
  seqnames_filter = NULL,
  genomic_ranges_filter = NULL,
  invert_granges_filter = FALSE,
  samples_subset = NULL
)
```

Arguments

meth_rse	A RangedSummarizedExperiment for methylation data.
n_sites	Number of sites to randomly sample. Default is 1000. Will give an error if there are less than this number of sites available to sample after applying any of the optional filters.
seqnames_filter	An optional character vector giving the names of sequences to filter meth_rse for.
genomic_ranges_filter	An optional GRanges object used to first subset meth_rse. Sites will then be chosen randomly from those overlapping these ranges.
invert_granges_filter	TRUE or FALSE indicating whether to invert the genomic_ranges_filter so as to exclude sites overlapping these regions. Default value is FALSE.
samples_subset	Optional sample names used to subset meth_rse.

Value

A RangedSummarizedExperiment with the specified number of randomly sampled sites after applying the different filters.

Examples

```
# Load sample RangedSummarizedExperiment with CpG methylation data
data(tubb6_meth_rse, package = "methodical")
tubb6_meth_rse <- eval(tubb6_meth_rse)

# Create a sample GRanges object to use to mask tubb6_meth_rse
mask_ranges <- GRanges("chr18:12305000-12310000")

# Get 20 random CpG sites outside mask_ranges
random_cpgs <- methodical::sampleMethSites(tubb6_meth_rse, n_sites = 20, genomic_ranges_filter = mask_ranges,
  invert_granges_filter = TRUE)

# Check that no CpGs overlap repeats
intersect(rowRanges(random_cpgs), mask_ranges)
```

strandedDistance	<i>Calculate distances of query GRanges upstream or downstream of subject GRanges</i>
------------------	---

Description

Upstream and downstream are relative to the strand of subject_gr. Unstranded regions are treated the same as regions on the "+" strand.

Usage

```
strandedDistance(query_gr, subject_gr)
```

Arguments

query_gr	A GRanges object
subject_gr	A GRanges object.

Value

A numeric vector of distances

Examples

```
# Create query and subject GRanges
query_gr <- GenomicRanges::GRanges(c("chr1:100-1000:+", "chr1:2000-3000:-"))
subject_gr <- GenomicRanges::GRanges(c("chr1:1500-1600:+", "chr1:4000-4500:-"))

# Calculate distances between query and subject
methodical::strandedDistance(query_gr, subject_gr)
```

```
summarizeRegionMethylation
```

Summarize methylation of genomic regions within samples

Description

Summarize methylation of genomic regions within samples

Usage

```
summarizeRegionMethylation(
  meth_rse,
  assay = 1,
  genomic_regions,
  genomic_region_names = NULL,
  col_summary_function = "colMeans2",
  keep_metadata_cols = FALSE,
  max_sites_per_chunk = floor(62500000/ncol(meth_rse)),
  na.rm = TRUE,
  BPPARAM = BiocParallel::SerialParam(),
  ...
)
```

Arguments

meth_rse	A RangedSummarizedExperiment with methylation values.
assay	The assay from meth_rse to extract values from. Should be either an index or the name of an assay. Default is the first assay.
genomic_regions	GRanges object with regions to summarize methylation values for.
genomic_region_names	A character vector of unique names to assign genomic_regions in the output table. Defaults to names(genomic_regions) if present or otherwise converts regions to character strings (e.g. "chr:1000-2000") to use as names.
col_summary_function	A function that summarizes column values. Should be the name of one of the column summary functions from MatrixGenerics. Default is "colMeans2".

keep_metadata_cols	TRUE or FALSE indicating whether to add the metadata columns of genomic_regions to the output. Default is FALSE.
max_sites_per_chunk	The approximate maximum number of methylation sites to try to load into memory at once. The actual number loaded may vary depending on the number of methylation sites overlapping each region, but so long as the size of any individual regions is not enormous ($\geq$ several MB), it should vary only very slightly. Some experimentation may be needed to choose an optimal value as low values will result in increased running time, while high values will result in a large memory footprint without much improvement in running time. Default is $\text{floor}(62500000/\text{ncol}(\text{meth_rse}))$, resulting in each chunk requiring approximately 500 MB of RAM.
na.rm	TRUE or FALSE indicating whether to remove NA values when calculating summaries. Default value is TRUE.
BPPARAM	A BiocParallelParam object. Defaults to <code>BiocParallel::SerialParam()</code> .
...	Additional arguments to be passed to <code>col_summary_function</code> .

Value

A data.table with the summary of methylation of each region in genomic_regions for each sample.

Examples

```
# Load sample RangedSummarizedExperiment with CpG methylation data
data(tubb6_meth_rse, package = "methodical")
tubb6_meth_rse <- eval(tubb6_meth_rse)

# Create a sample GRanges
test_gr <- GRanges(c("chr18:12303400-12303500", "chr18:12303600-12303750", "chr18:12304000-12306000"))
names(test_gr) <- paste("region", 1:3, sep = "_")

# Calculate mean methylation values for regions in test_gr
test_gr_methylation <- methodical::summarizeRegionMethylation(tubb6_meth_rse, genomic_regions = test_gr,
  genomic_region_names = names(test_gr))
```

sumTranscriptValuesForGenes

Combine the expression values of transcripts to get overall expression of their associated genes

Description

Combine the expression values of transcripts to get overall expression of their associated genes

Usage

```
sumTranscriptValuesForGenes(  
  transcript_expression_table,  
  gene_to_transcript_list  
)
```

Arguments

transcript_expression_table

A table where rows are transcripts and columns are samples. Row names should be the names of transcripts.

gene_to_transcript_list

A list of vectors where the name of each list entry is a gene name and its elements are the names of transcripts. Can alternatively be a GRangeList where the name of each list element is a gene and the names of the individual ranges are transcripts.

Value

A data.frame with the sum of transcript expression values for genes where rows are genes and columns are samples

tubb6_correlation_plot

tubb6_correlation_plot

Description

A plot of the correlation values between methylation-transcription correlations for CpG sites around the TUBB6 TSS.

Usage

tubb6_correlation_plot

Format

A ggplot object.

tubb6_cpg_meth_transcript_cors
<i>tubb6_cpg_meth_transcript_cors</i>

Description

A data.frame with the methylation-transcription correlation results for CpGs around the TUBB6 TSS.

A data.frame with the correlation results for CpG sites within +/- 5 KB of the TUBB6 (ENST00000591909) TSS.

Usage

tubb6_cpg_meth_transcript_cors

tubb6_cpg_meth_transcript_cors

Format

A ggplot object.

A data.frame with 5 columns giving the name of the CpG site (meth_site), name of the transcript associated with the TSS, Spearman correlation value between the methylation of the CpG site and expression of the transcript, p-value associated with the correlations and distance from the CpG site to the TSS.

tubb6_meth_rse	<i>tubb6_meth_rse</i>
----------------	-----------------------

Description

The location of the TSS for TUBB6.

Usage

tubb6_meth_rse

Format

A call to create a RangedSummarizedExperiment with methylation data for 355 CpG sites within +/- 5,000 base pairs of the TUBB6 TSS in 126 normal prostate samples. Should be evaluated after loading using `tubb6_meth_rse <- tubb6_meth_rse <- eval(tubb6_meth_rse)` to restore the RangedSummarizedExperiment.

Source

WGBS data from 'Li, Jing, et al. "A genomic and epigenomic atlas of prostate cancer in Asian populations." Nature 580.7801 (2020): 93-99.'

tubb6_tmrs	<i>tubb6_tmrs</i>
------------	-------------------

Description

TMRs identified for TUBB6

Usage

tubb6_tmrs

Format

A GRanges object with two ranges.

tubb6_transcript_counts	<i>tubb6_transcript_counts</i>
-------------------------	--------------------------------

Description

Transcript counts for TUBB6 in normal prostate samples.

Usage

tubb6_transcript_counts

Format

A data.frame with normalized transcript counts for TUBB6 in 126 normal prostate samples.

Source

RNA-seq data from 'Li, Jing, et al. "A genomic and epigenomic atlas of prostate cancer in Asian populations." Nature 580.7801 (2020): 93-99.'

tubb6_tss	<i>tubb6_tss</i>
-----------	------------------

Description

The location of the TSS for TUBB6.

Usage

tubb6_tss

Format

GRanges object with 1 ranges for the TUBB6 TSS.

Source

The TSS of the ENST00000591909 transcript.

TumourMethDatasets	<i>TumourMethDatasets</i>
--------------------	---------------------------

Description

A table describing the datasets available from TumourMethData.

Usage

TumourMethDatasets

Format

A data.frame with one row for each dataset

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